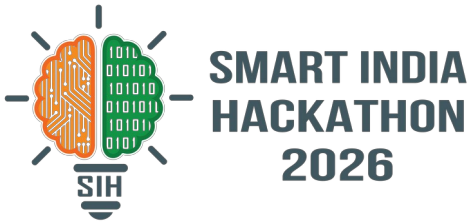


SMART INDIA HACKATHON 2026



RASTA AI **SIH26002**

SHORTEST IS NOT ALWAYS
USABLE.

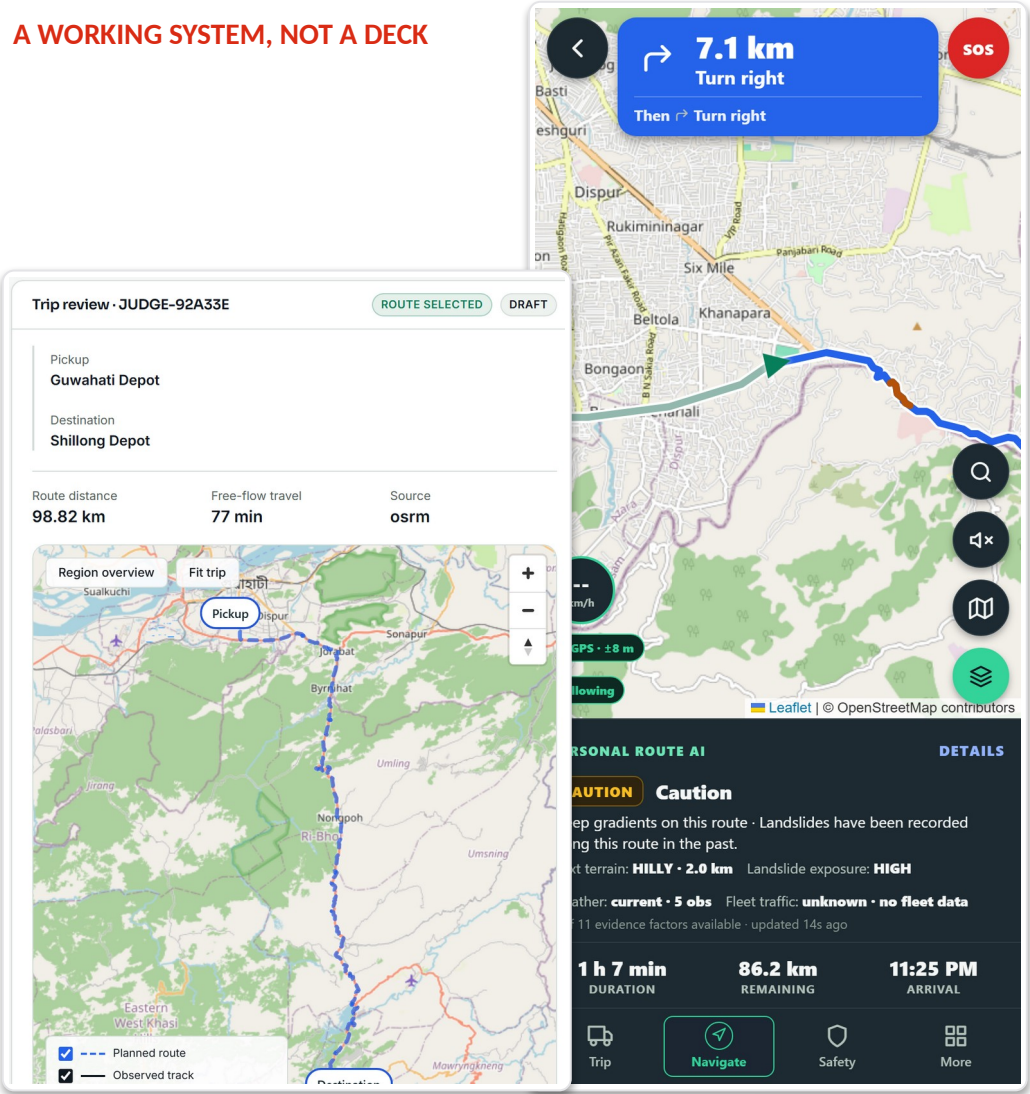
“Can this essential-supply truck reliably use this corridor now?”

PROBLEM STATEMENT	SIH26002 · MDoNER
TITLE	AI-Based Smart Logistics and Accessibility Intelligence Platform (NER)
THEME / CATEGORY	Smart Automation · Software
TEAM	NER-AI LOGISTICS · Team ID 17 (internal group no.)

Route Intelligence for Essential-Supply Logistics in North Eastern India

End-to-end working prototype · physical-phone validated · live demo corridor Guwahati → Shillong

A WORKING SYSTEM, NOT A DECK



A road can be the shortest and still be **unusable**.

NER corridors close without notice: monsoon rain, landslides, floods, official warnings, single-lane hill roads, no phone signal. A shortest-path router still sends the truck.

A NORMAL ROUTER

Origin → Shortest / fastest road → Truck

No governance. No answer when data is missing.

RASTA AI



Manager governance → driver navigation, 23 languages, offline package

UNKNOWN ≠ SAFE

Missing or stale evidence is marked UNKNOWN, never counted as safe, and the route goes to an authorised reviewer.

MANAGER · CHECK CONDITIONS

REQUIRES REVIEW

Hazard evidence is incomplete or elevated. An authorised reviewer must review this route before selection. Refresh conditions after review.

Landslide data is not available - no source is connected. · Steep gradients on this route · Landslides have been recorded along this route in the past. · The landslide inventory ends in 2017 - recent years are not covered. · River levels near this route are close to their recent average. · No official disaster alert names a district on this route right now. · No fleet truck has driven this road in the last fifteen minutes, so traffic is unknown.

Unavailable: Fuel range, Landslide, Road quality, Truck restrictions

Evidence coverage · Weather AVAILABLE · Terrain AVAILABLE · Warnings AVAILABLE · Traffic UNKNOWN

TERRAIN
0.5 KM STEEP
49-1436 m · 1975 m climb · steepest 10.3%
flat 61.8 km · rolling 29.6 km · hilly 7.0 km · steep 0.5 km
Copernicus DEM GLO-90 via Open-Meteo + SRTM 30 m via OpenTopoData · 100% of 198 samples answered

LANDSLIDE HISTORY
HIGH EXPOSURE
22 recorded landslides within 5 km of the road, nearest 0.4 km
6 more nearby placed too imprecisely to count · HIGH is three or more; MODERATE is one or two
NASA Global Landslide Catalog · inventory 2007-2017 — aged, recent years not covered

RIVER LEVELS
NORMAL
Discharge at 5 river cells along the corridor, highest 1.0x its own 30-day mean · GloFAS via Open-Meteo, 2026-09-14. A level, not a flood forecast and not a road-closure claim.

OFFICIAL ALERTS
NONE ON CORRIDOR
NDMA SACHET CAP feed, 99 alerts nationwide at 10:13:54 PM · placed by district name (Dispur, East Khasi Hills, Kamrup Metropolitan, Myllem, Ri-Bhoi, Umling, Umsning) · 4 more active elsewhere in the corridor states, not placeable on this road.

FLEET TRAFFIC
UNKNOWN

Real screen · evidence · freshness · UNKNOWN

Evidence decides. AI explains, and never overrules.

WEATHER Open-Meteo · MET Norway	TERRAIN Copernicus DEM · OpenTopoData	WARNINGS NDMA SACHET (CAP)	FLOOD GloFAS	LANDSLIDE HISTORY NASA GLC 2007–17	TRAFFIC fleet GPS fixes
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ROUTE-SPECIFIC EVIDENCE · FastAPI backend

every factor sampled along the selected road · PostgreSQL + PostGIS (Supabase) · OSRM routes, alternatives, turn steps

DETERMINISTIC SAFETY POLICY

11 factors · reason codes · UNKNOWN ≠ SAFE · reviewer authorisation when hazard data is missing

CONTINUE

CAUTION

HOLD

REROUTE

MANAGER REVIEW

DRIVER NAVIGATION

ROUTE-AHEAD 60 s

CONDITIONS CHANGE?

REASSESS

CLIENTS Manager console — React + TypeScript, manager accounts only · Driver app — React Native / Expo, Android; one login, server-decided role, offline trip package

STACK FastAPI · PostgreSQL / PostGIS (Supabase) · React · React Native / Expo · OSRM · OpenStreetMap · Gemini

WHO DECIDES

DETERMINISTIC SAFETY ENGINE

makes every operational decision

GEMINI / OPENROUTER (LLM)

explains the decision in the driver's language — never decides

EXPERIMENTAL ML

research channel only; held outside routing by the safety gate

GOVERNANCE CHAIN

Verified evidence



Deterministic safety policy



Manager governance



Driver action

It already runs on a real phone, on a real road.

The whole lifecycle below was run end to end on a physical Android phone against the hosted backend.

12/12 physical judge-flow steps

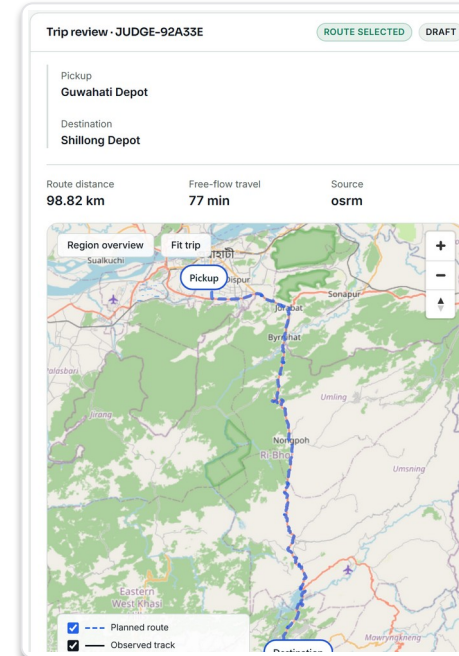
9/9 physical role-flow steps

1138+ backend · **621+** driver · **170+** manager tests

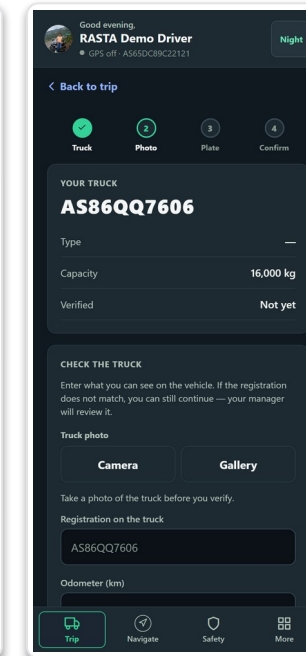
PHYSICAL ANDROID · CERTIFIED

Canonical demo Guwahati → Shillong · ≈ 98.8 km real route

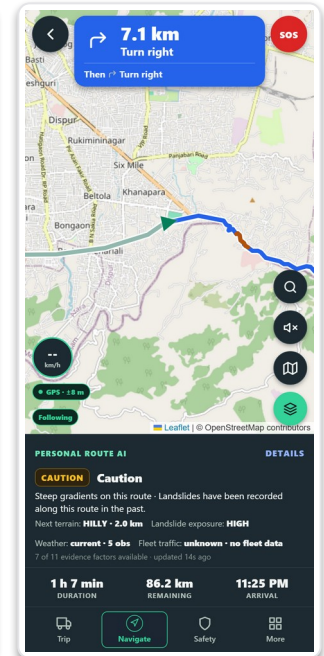
CERTIFIED LIFECYCLE



Manager · route review



Driver · truck check



Driver · navigation

KNOWN LIMITS → MITIGATION Render cold start → warm /health before the demo · provider rate limits → cached evidence + explicit UNKNOWN · background push needs Firebase config → in-app alerts already work · experimental ML → outside routing until the gate passes

Food, medicine, fuel and relief reach people when the corridor is decided on evidence.

WHO BENEFITS

FLEET MANAGERS

Route usability and uncertainty visible before dispatch; auditable decisions.

DRIVERS

Safer navigation, danger context, emergency support, 23 languages.

REMOTE COMMUNITIES

More resilient access to food, medicine, fuel and relief supplies.

GOVERNMENT / OPERATIONS

Auditable route evidence and disruption visibility per corridor.

DEMONSTRATED TODAY

23 selectable languages, explicit fallback status

11 route factors in every decision

60 s route-ahead monitor while driving

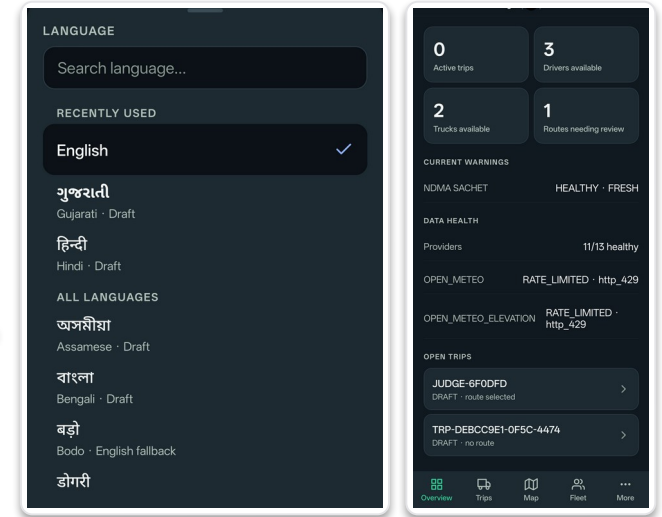
12/12 judge-flow steps certified on a physical phone

100 % of reroutes reviewed by a human

0 unknown factors counted as safe

Verified evidence → deterministic safety policy → manager governance → driver action. AI has no uncontrolled authority; no invented percentages.

REAL SCREENS • APK 1.0.18



Driver • 23 languages

Manager • mobile

Real screens, APK 1.0.18. One login, server-decided role: a manager gets a mobile fleet view, a driver gets navigation. Each language shows its status — Verified, Draft or English fallback — so nobody is misled.

We trained it, tested it, and **refused to ship it.**

RESEARCH / EVIDENCE SOURCES BEHIND THE ROUTE DECISION

ROUTING / PLACES

OSRM • OpenStreetMap • Nominatim

real road geometry, alternatives and turn steps; place search on the corridor

OFFICIAL / FLOOD

NDMA SACHET (CAP) • Copernicus GloFAS

official warnings polled with freshness; river discharge against its 30-day mean

AI ASSISTANCE

Google Gemini

explains the decision in the driver's language — never safety authority

WEATHER / TERRAIN

Open-Meteo • MET Norway • Copernicus DEM • OpenTopoData

forecast sampled along the road; climb, gradient and steep segments (GLO-90)

LANDSLIDE HISTORY

NASA Global Landslide Catalog

historical exposure; inventory 2007–2017, ≤5 km location-accuracy filter

PROBLEM STATEMENT

MDoNER • SIH26002

AI-based smart logistics and accessibility intelligence for the NER

EXPERIMENTAL LANDSLIDE RESEARCH

95 %

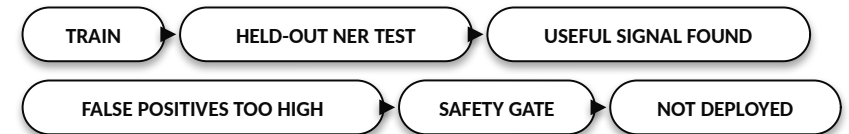
RECALL

on the NER geographic holdout — 154 catalogued events, no NER row used in training

SAFETY GATE

NOT DEPLOYED

FPR 50 % too high for production deployment



VALIDATION BEFORE AUTOMATION. Strong recall was not enough. Held-out testing exposed the false-positive rate, so the gate keeps the model out of routing.

Precision 0.32 • F1 0.48 • PR-AUC 0.55 • ROC-AUC 0.83 • Brier 0.12 • calibrated logistic regression on NASA GLC + ERA5-Land rain + DEM slope • docs/HAZARD_ERROR_ANALYSIS.md

REFERENCES gpm.nasa.gov/landslides (NASA GLC) • sachet.ndma.gov.in • global-flood.emergency.copernicus.eu (GloFAS) • open-meteo.com • api.met.no • opentopodata.org • Copernicus DEM • project-osrm.org • openstreetmap.org • github.com/nxtlucifer/ner-ai-logistics (repository, tests, model registry)